Multi-View Diffusion under Test-Time Scaling

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Abstract

Test-time scaling (TTS) improves generative performance by allocating additional inference computation, yet its effectiveness for structured outputs such as multi-view diffusion models remains underexplored. In this work, we systematically investigate how generation, verification, and search interact when the generated views must remain consistent. We first find that joint multi-view generation exhibits stronger Oracle scaling than independent per-view generation across multiple evaluated models, while revealing a fundamental tension between preserving cross-view consistency through set-level selection and improving individual views through view-level selection. To address this challenge, we introduce DINO-centroid, a lightweight verifier that leverages cross-sample consensus for complete-set selection and uncertain-view identification. Our analysis shows that DINO-centroid becomes increasingly aligned with the ground-truth representation as the sampling budget grows, suggesting that the verifier itself exhibits a form of scaling behavior absent from conventional TTS verifiers. Building on these findings, we propose Exploration–Refinement Search (E-R Search), a strategy that combines set-level exploration with targeted view-level regeneration by formulating the latter as a diffusion inverse problem. Under matched inference budgets, our strategy achieves higher reconstruction quality than existing test-time search baselines. Our findings highlight the importance of accounting for component dependencies in generation, verification, and search for effective TTS of structured outputs.

1 Introduction

Test-time scaling (TTS) can improve the task performance of diffusion models by allocating additional computation at inference time. Several approaches rank complete candidate outputs or search their denoising trajectories, with applications spanning image synthesis (Ma et al., 2025; Ramesh & Mardani, 2025), planning (Zhang et al., 2025), and constraint-based reasoning (Zhang et al., 2026). Beyond treating each generated output as an indivisible whole, recent approaches recognize that an output may consist of multiple interdependent components and exploit this structure through autoregressive continuation (Liu et al., 2025) and conditional region regeneration (Kang et al., 2026). They motivate a broader question: how should the choice of generation mechanism, verification granularity, and search strategy reflect dependencies among output components?

To address this question, we systematically investigate TTS for structured output spaces, where each sample comprises components that are statistically dependent under the target distribution. A representative example is **single-image-conditioned multi-view generation**, a key component of modern image-to-3D pipelines, where each output consists of a set of interdependent views of the same object. A single reference image does not uniquely determine the object's unseen geometry and appearance, leaving multiple plausible hypotheses for the target views. This ambiguity creates opportunities for TTS to explore alternative hypotheses.

To understand how to exploit these opportunities, we organize our investigation around three research questions. **Generation:** How does TTS behave under independent component generation and joint generation that explicitly models dependencies? **Verification:** Which signals enable reliable evaluation without

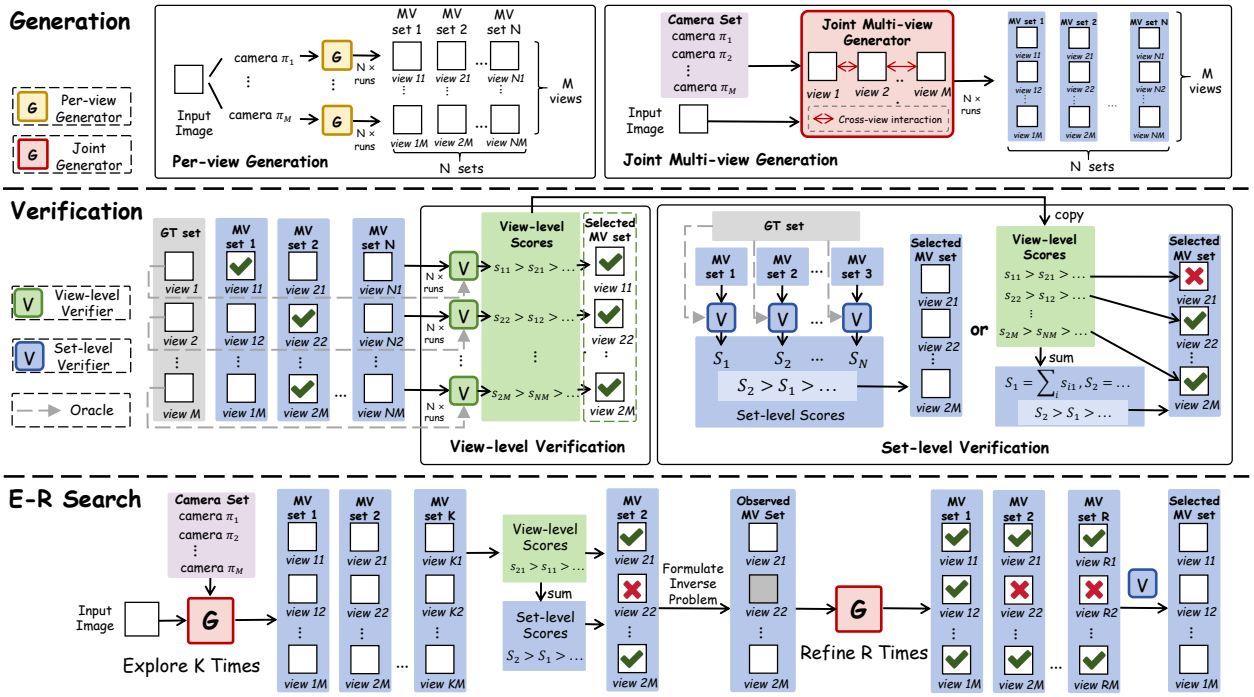


Figure 1: **Overview of multi-view generation, verification, and search. Row 1: Generation.** Per-view generation synthesizes viewpoints independently, while joint generation models cross-view dependencies. **Row 2: Verification.** View-level verification independently chooses the best candidate for each viewpoint, whereas set-level verification preserves complete multi-view samples. Oracle verification, when available, uses the corresponding ground-truth targets. **Row 3: E-R Search.** Our E-R Search generates K candidate sets and performs R refinement trajectories, preserving reliable views while regenerating uncertain ones.

ground truth, and how should component-level scores and complete-sample scores guide selection? **Search:** How should a fixed inference budget be divided between exploring new complete samples and conditionally refining unreliable components of an existing sample? These questions are interconnected. A joint generator can produce correlated components, yet independently selecting components from different candidates may disrupt those relationships. Conversely, component-level scores can support complete-sample selection and identify where further refinement is needed.

Our generation study compares TTS under independent per-view and joint multi-view generation. Across multiple evaluated models, we find larger scaling gains for the joint generator as the sampling budget increases, motivating our focus on joint generation. Under practical verification, set-level selection outperforms independent view composition and better preserves cross-view consistency, while Oracle view-wise selection reveals additional room for local improvement. These observations expose a tension between retaining the dependencies established during generation and exploiting improvement opportunities at individual viewpoints.

To address this tension, we propose a novel lightweight verifier, termed **DINO-centroid**, which aggregates self-supervised visual representations across independent generations at each viewpoint to construct view-specific consensus references for scoring candidate views (Caron et al., 2021; Oquab et al., 2023). These component scores can be aggregated to select a complete set and used individually to identify views for refinement, separating scoring granularity from selection granularity. DINO-centroid achieves a favorable balance between selection quality and computational cost. **Moreover, its consensus reference becomes closer to the ground truth as the sampling budget grows, suggesting that the verifier itself**

exhibits a form of scaling behavior, a property absent from conventional TTS verifiers.

Our search study examines whether additional computation is better spent on independent exploration or conditional refinement. Controlled experiments first show that providing additional ground-truth views as observations improves reconstruction of the remaining viewpoints, motivating the reuse of reliable generated content. We then propose a novel search strategy, termed **Exploration–Refinement Search (E-R Search)**, which combines global exploration with targeted view-level refinement. E-R Search generates multiple complete multi-view samples, selects a promising set using DINO-centroid, and identifies its uncertain views for conditional regeneration. By treating the retained reliable views as observations, we formulate refinement as a diffusion inverse problem (Chung et al., 2023) and solve it using Posterior-DDIM (Jiao et al., 2026). Under matched sampling budgets, this strategy achieves higher reconstruction quality than both classical and recent search baselines. We further examine how the allocation between exploration and refinement, and the strength of observation conditioning, affect performance.

Our study provides empirical guidance for TTS in multi-view generation and identifies principles for exploiting component structure in TTS. Our main contributions are threefold:

- We present the first systematic study of TTS for multi-view diffusion models, investigating how generation paradigms, verification granularity, and search strategies interact under cross-view dependencies.
- We propose **DINO-centroid**, a novel lightweight verifier that leverages cross-sample feature consensus for complete-set selection and uncertain-view identification. We further show that its consensus reference becomes more accurate as the sampling budget increases, which is a desirable property largely absent from existing verification signals.
- We propose **Exploration–Refinement Search (E-R Search)**, a novel search strategy that combines global exploration with targeted view-level regeneration by formulating refinement as a diffusion inverse problem. Under matched computational budgets, E-R Search achieves higher reconstruction quality than existing test-time search baselines.

2 Background

Fig. 1 provides an overview of multi-view generation, verification, and search, which we introduce in the following. Given an input image c and predefined camera poses $\Pi = \{\pi_v\}_{v=1}^{M}$, the goal is to generate a complete multi-view set $\mathbf{X} = (x^{(1)}, \dots, x^{(M)})$, where $x^{(v)}$ denotes the image rendered from camera pose π_v. Multi-view generation can follow two paradigms: *per-view generation*, which independently generates each target view, and *joint multi-view generation*, which models dependencies among viewpoints. Their corresponding distributions can be expressed as

$$\text{Per-view: } x^{(v)} \sim p_\theta(x^{(v)} \mid c, \pi_v); \qquad \text{Joint: } \mathbf{X} \sim p_\theta(\mathbf{X} \mid c, \Pi). \tag{1}$$

For TTS, we generate N independent multi-view sets $\{\mathbf{X}_i\}_{i=1}^{N}$. **We distinguish two verification strategies based on the granularity of selection.** In *set-level verification*, a complete multi-view set is selected as a whole, whereas in *view-level verification*, each viewpoint is selected independently across sets. Importantly, **verification granularity does not necessarily coincide with scoring granularity**: a view-level verifier, such as Aesthetic Score, can also support set-level verification by averaging its per-view scores. The two strategies are formally defined as

$$\text{Set-level: } i^* = \arg\max_i R_{\text{set}}(\mathbf{X}_i); \qquad \text{View-level: } i_v^* = \arg\max_i R_{\text{view}}(x_i^{(v)}). \tag{2}$$

In Fig. 1, we use the shorthand $S_i = R_{\text{set}}(\mathbf{X}_i)$ for the set-level score, and $s_{iv} = R_{\text{view}}(x_i^{(v)})$ for the view-level score. Set-level verification retains the complete set $\mathbf{X}^* = \mathbf{X}_{i^*}$, whereas view-level verification composes independently selected views, $\mathbf{X}^* = (x_{i_1^*}^{(1)}, \ldots, x_{i_M^*}^{(M)})$. Set-level verification selects a complete multi-view set, preserving the cross-view dependencies established by joint generation, but may still retain locally suboptimal views.

Diffusion models can serve as generative priors for inverse problems, recovering unknown content from partial observations. Finally, for search, as shown in Row 3 of Fig. 1, we formulate uncertain-view regeneration as an inverse problem by treating reliable views as observations under the joint multi-view diffusion prior.

Multi-view Diffusion Models. Diffusion-based multi-view generation can be broadly categorized into *per-view generation* and *joint multi-view generation*. Per-view methods, such as Zero-1-to-3 (Liu et al., 2023) and 3DiM (Watson et al., 2022), synthesize target viewpoints individually using view-conditioned diffusion models. Related approaches, such as SparseFusion (Zhou & Tulsiani, 2023), further leverage view-conditioned diffusion priors for sparse-view 3D reconstruction.

Joint multi-view methods instead generate multiple views through a shared denoising process. SyncDreamer (Liu et al., 2024) synchronizes multi-view generation through 3D-aware cross-view interactions, while Zero123++ (Shi et al., 2023) generates consistent multi-view images using image-conditioned diffusion. Subsequent approaches further improve multi-view generation through cross-domain diffusion, image-prompt conditioning, and efficient cross-view interactions (Long et al., 2024; Wang & Shi, 2023; Li et al., 2024; Huang et al., 2025). SV3D (Voleti et al., 2024) adapts video diffusion for camera-controlled multi-view synthesis, while MVDream (Shi et al., 2024) explores joint multi-view diffusion for text-conditioned 3D generation. Multi-view diffusion models also serve as key components of modern image-to-3D pipelines (Wu et al., 2024; Wang et al., 2024; Xu et al., 2024).

Test-Time Scaling for Diffusion Models. Test-time scaling (TTS) improves generative performance by allocating additional inference computation without modifying model parameters. Ma et al. (2025) investigate verifier-guided search over diffusion sampling noise. Subsequent studies explore noise trajectory optimization (Ramesh & Mardani, 2025), direct noise optimization (Tang et al., 2025), and local and global search (Zhang et al., 2025). Other approaches employ sequential Monte Carlo and reward-guided particle resampling (Kim et al., 2025; Singhal et al., 2025), step-wise prompt refinement and candidate selection (Qiao et al., 2025), or iterative reward-guided refinement (Uehara et al., 2025), and verifier-free reasoning (Zhang et al., 2026).

Recent studies exploit the internal structure of generated outputs. Video-T1 (Liu et al., 2025) performs adaptive branching and pruning along autoregressive video generation, while Iterative Partial Refinement (Kang et al., 2026) randomly regenerates subsets of regions without external verification. In contrast, we study verifier-guided TTS for diffusion models with interdependent output components.

Diffusion Models for Inverse Problems. Diffusion models have been widely used as generative priors for inverse problems, including image restoration and reconstruction. DDRM (Kawar et al., 2022) addresses linear inverse problems, while DDNM (Wang et al., 2023) exploits the null space of the measurement operator to preserve observation consistency. DPS (Chung et al., 2023) extends posterior sampling to general noisy inverse problems, and Posterior-DDIM (Jiao et al., 2026) introduces SNR-dependent updates for efficient posterior sampling.

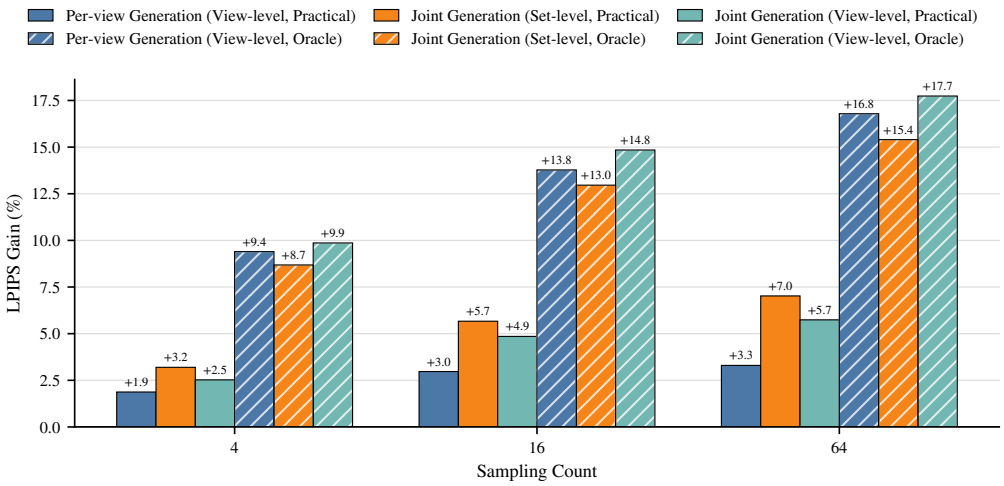


Figure 2: **Per-view vs. joint multi-view generation under TTS.** Joint generation (instantiated by SyncDreamer) exhibits stronger Oracle scaling than per-view generation (instantiated by Zero-1-to-3). For SyncDreamer, set-level selection performs better under practical verification, while view-level selection is stronger under Oracle verification.

3 Systematic Investigation of Multi-view TTS

We begin with pilot experiments (Sec. 3.1) that motivate and guide our investigation of generation (Sec. 3.2), verification (Sec. 3.3), and search (Sec. 3.4).

3.1 Motivation

Ambiguity and Test-Time Scaling Opportunities. Single-image-conditioned multi-view generation is inherently ambiguous, as a single observation does not uniquely determine the geometry, appearance, and occluded content of the object. As a result, different stochastic generations conditioned on the same reference image may produce distinct yet plausible multi-view hypotheses (see Fig. 11 in Appendix C). This ambiguity may offer greater scope for test-time scaling than in less ambiguous generation tasks, as additional samples can explore alternative hypotheses about unobserved views. This motivates our investigation of how generation paradigms affect scaling behavior.

Component Dependencies and Selection Granularity. Across evaluated models in Sec. 3.2, joint multi-view generation, which explicitly models dependencies across viewpoints, exhibits stronger scaling behavior than independent per-view generation. At the same time, these cross-view dependencies introduce a challenge in the granularity of verification and selection. Independently selecting views from different sets may disrupt their joint structure and compromise cross-view consistency, whereas selecting an entire multi-view set preserves these dependencies but may leave locally suboptimal views uncorrected. Fig. 2 further reveals this tension: for joint generation, under practical verification, set-level selection outperforms view-level selection, whereas the opposite holds under Oracle verification. These observations motivate our investigation of verification granularity and search strategies that allocate a fixed inference budget between independent exploration and targeted view-level refinement.

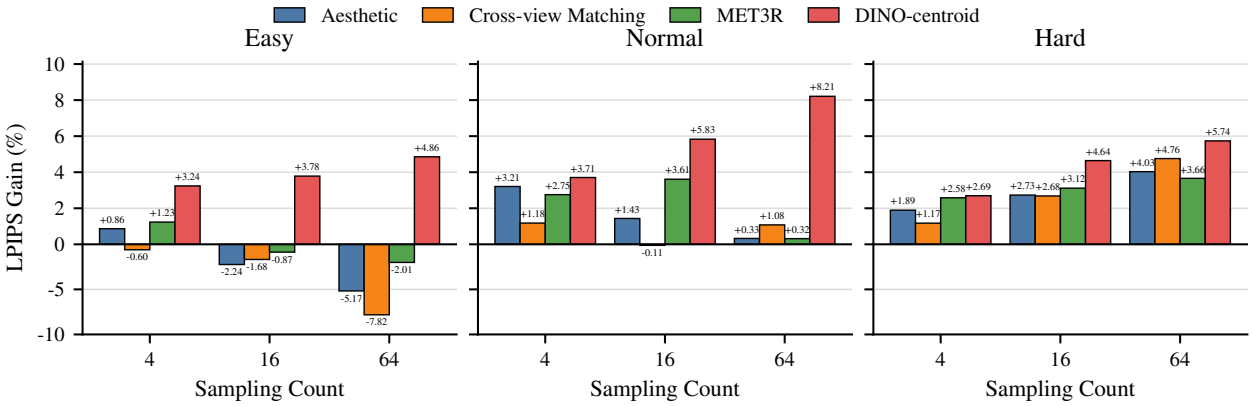


Figure 3: **Best-of-N performance using different verifiers.** DINO-centroid achieves a favorable balance between selection quality and verification cost across sampling budgets and difficulty levels.

3.2 Generation: Independent vs. Joint Generation

Our first investigation examines how generation paradigms affect test-time scaling. We compare Zero-1-to-3 (Liu et al., 2023), a representative per-view generation model that generates target views independently, with SyncDreamer (Liu et al., 2024), a representative joint multi-view generation model. Under the same view-level Oracle Best-of-N selection for $N \in \{4, 16, 64\}$, SyncDreamer exhibits larger scaling gains in reconstruction quality, as shown in Fig. 2. This motivates our focus on joint generation in the subsequent verification and search studies. To further validate this trend, we provide additional scaling comparisons between representative per-view and joint generation models in Appendix H.

3.3 Verification: Signals and Selection Granularity

Our second investigation examines which verification signals can reliably evaluate multi-view generations without ground-truth target views and how verification granularity affects selection.

Verification signals. We evaluate a range of verifiers for multi-view generation. The Oracle verifier evaluates reconstruction quality using ground-truth LPIPS. Aesthetic Score (Schuhmann et al., 2022) measures perceptual image quality, while Cross-view Matching (Edstedt et al., 2024) and MEt3R (Asim et al., 2025) assess cross-view consistency through 2D correspondences and geometric compatibility, respectively. Further details and computational costs are summarized in Appendix D.2, Table 1.

In contrast, our proposed DINO-centroid verifier leverages cross-sample feature consensus without explicit 3D reconstruction or dense correspondence estimation. Specifically, we construct a view-specific centroid by averaging and normalizing DINO features (Caron et al., 2021; Oquab et al., 2023) across independent generations. Each candidate view is scored by its cosine similarity to the corresponding centroid, and these scores are averaged to obtain a set-level score. We use the latter to select a complete multi-view set and the former to identify suboptimal views for subsequent refinement. The mathematical formulation and implementation details are provided in Appendix A.

We evaluate the verifiers through Best-of-N selection under matched sampling budgets. As shown in Fig. 3, DINO-centroid consistently achieves the strongest selection performance among the evaluated practical verifiers across sampling budgets and difficulty levels. It also incurs the lowest verification cost in Table 1, providing a favorable balance between selection quality and efficiency.

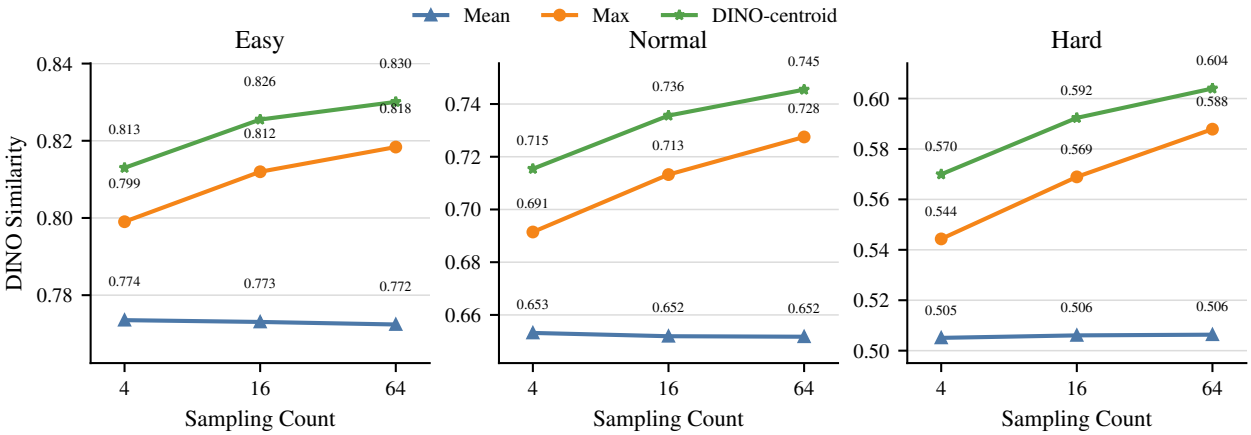


Figure 4: **Cross-sample consensus in DINO feature space.** We compare the ground-truth similarity of individual candidates (the mean and max of their similarity values) with that of the corresponding view-specific centroid. The centroid provides a more accurate consensus representation under the evaluated settings, and its similarity improves with the sampling budget.

Why does DINO-based consensus work? We first compute DINO cosine similarity, PSNR, SSIM, and LPIPS between each generated view and its corresponding ground-truth view, and measure the correlation between DINO similarity and the reconstruction metrics. DINO similarity shows moderate correlations with reconstruction metrics, supporting its use for candidate ranking. Detailed correlation analysis is provided in Appendix B.

We further investigate whether DINO-centroid yields a more accurate representation of the underlying object. As shown in Fig. 4, DINO-centroid is consistently closer to the ground-truth representation than individual candidates, and its similarity further improves as the sampling budget increases, particularly on the Normal and Hard subsets. This suggests that **DINO-centroid benefits directly from additional samples, exhibiting a form of verifier scaling that conventional TTS verifiers typically do not exhibit.**

Selection granularity. Although DINO-centroid provides view-level scores, view-level selection may combine incompatible hypotheses and disrupt cross-view consistency. We compare set-level selection with view-level selection using the same candidate pool and verification signal. Fig. 8 shows that set-level selection (BoN) consistently outperforms view-level selection (Composed-view) in reconstruction quality under DINO-centroid signals. We further evaluate the consistency difference using MEt3R. Fig. 5 shows that independent view composition degrades cross-view consistency, as evidenced by consistently positive mean $\Delta\mathrm{MEt3R}$ values and a majority of objects exhibiting worse consistency, particularly for harder objects.

Interestingly, the trend reverses under Oracle verification: Fig. 7 shows that view-level selection (Composed-view) outperforms set-level selection (BoN) in reconstruction quality when ground-truth information is available. This reveals that even the best complete sample may contain locally suboptimal views. Together, these findings expose a fundamental tension between preserving cross-view consistency through set-level selection and exploiting opportunities for view-level improvement, motivating our investigation of targeted refinement.

3.4 Search: Exploration and Refinement

Our third investigation examines how a fixed inference budget should be allocated between independent exploration and targeted view-level refinement. We compare representative search strategies and investigate

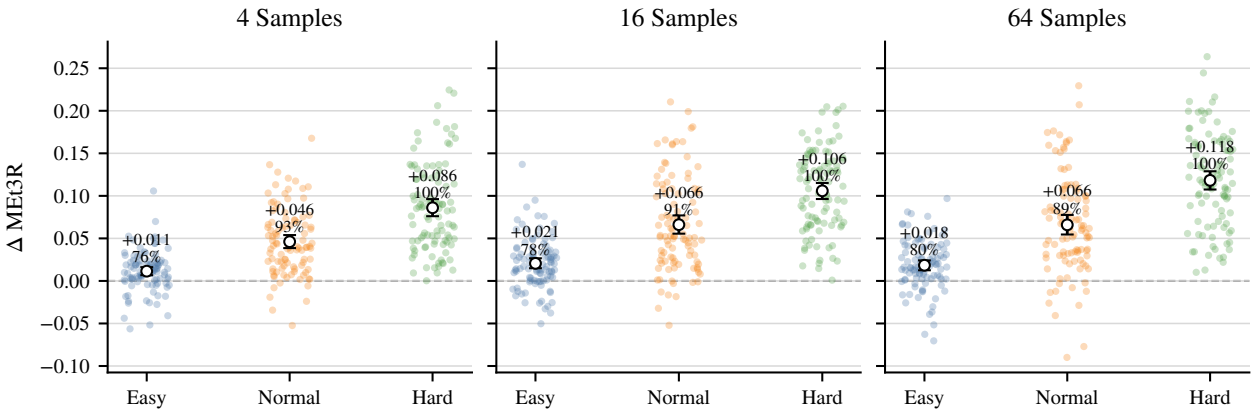


Figure 5: **Cross-view consistency under different selection granularities.** MEt3R ranges from 0 to 1, with lower values indicating higher cross-view consistency. We report $\Delta\text{MEt3R} = \text{MEt3R}_{\text{view}} - \text{MEt3R}_{\text{set}}$ across Easy, Normal, and Hard subsets. Positive values indicate worse consistency under independent view composition. Numbers denote the mean difference and the percentage of objects with positive differences.

whether reliable views can guide the regeneration of uncertain views. Building on these findings, we propose Exploration–Refinement Search (E-R Search) and evaluate it under both Oracle and practical verification.

Can reliable observations improve regeneration? Before introducing E-R Search, we conduct a controlled experiment to examine whether additional observations can improve the reconstruction of unobserved viewpoints. We provide different numbers of ground-truth views as fixed observations during Posterior-DDIM sampling and reconstruct the remaining views. Reconstruction metrics are computed only over the unobserved viewpoints, excluding the input view and all provided ground-truth observations. As shown in Fig. 6, a single refinement trajectory reduces LPIPS, with larger improvements as more ground-truth views are provided. Notably, when the ground-truth observations are replaced with reliable views identified by DINO-centroid from generated multi-view samples, refinement yields only about a 3% improvement, roughly an order of magnitude smaller than that achieved using seven ground-truth observations. This substantial gap suggests considerable potential for refinement when more reliable observations are available. These findings motivate a search strategy that exploits reliable views from previously generated samples to guide the regeneration of uncertain viewpoints. They also reveal a close connection between **generation**, **verification**, and **search**: more accurate verification can more reliably distinguish reliable and unreliable generated views, while improvements in the underlying generator can increase the quality of the generated views themselves. Both can provide stronger guidance for view-level refinement, suggesting that the effectiveness of such a search strategy may further improve as generation and verification capabilities advance.

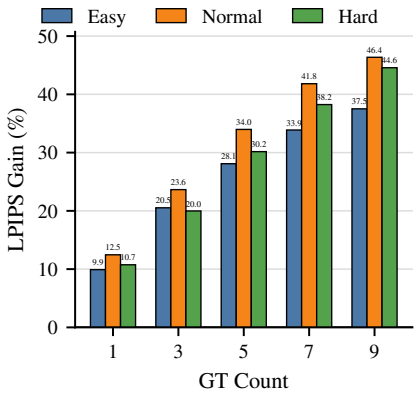


Figure 6: **Performance with different numbers of ground-truth (GT) observations.**

Search algorithms and budget setting. We compare five strategies under the same nominal sampling budgets $N \in \{4, 16, 64\}$. **Best-of-N** selects the highest-scoring complete sample from N independent generations. **View-composed Best-of-N** independently selects the highest-scoring view at each viewpoint from the same candidate pool. **Beam Search** progressively selects and expands promising intermediate

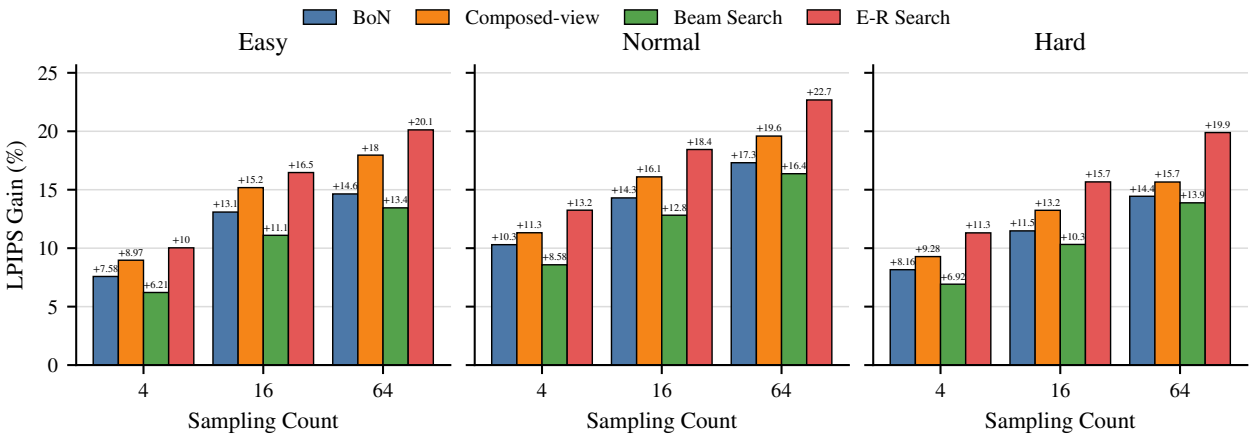


Figure 7: **Search performance under Oracle verification.** For each search strategy, the best candidate is selected based on LPIPS to the corresponding ground-truth.

diffusion states. **IPR-style Refinement** is our multi-view adaptation of the verifier-free partial-refinement principle in IPR. While IPR (Kang et al., 2026) randomly regenerates partial regions of an output, we instantiate this idea at the view level by randomly selecting target views for conditional regeneration while preserving the remaining views. **Exploration–Refinement Search (E-R Search)** allocates the same budget between independent exploration and verifier-guided conditional refinement of uncertain views. Detailed search configurations, including IPR-style Refinement, are provided in Appendix G.4 and Appendix G.5.

Exploration–Refinement Search. We propose a novel search strategy, termed **Exploration–Refinement Search (E-R Search)**, which combines global exploration with targeted view-level refinement. E-R Search first generates K multi-view sets and selects one using the set-level DINO-centroid score. Within the selected set, a view is considered *reliable* if it has the highest DINO-centroid score among the K candidates at the same viewpoint, and *uncertain* otherwise. We retain reliable views as observations and formulate the regeneration of uncertain views as a diffusion inverse problem under the joint multi-view prior. We then solve this inverse problem using Posterior-DDIM (Jiao et al., 2026). We set $K + R = N$ to match the sampling budget of Best-of-N, where R denotes the number of refinement trajectories. Details are provided in Appendix E.

Search performance under Oracle verification. We evaluate search strategies using ground-truth metrics for candidate selection, without providing ground-truth observations during refinement. As shown in Fig. 7, view-composed Best-of-N (i.e., Composed-view) outperforms conventional Best-of-N (i.e., BoN), revealing opportunities for view-level improvement. E-R Search further improves upon view composition under matched sampling budgets by generating new views conditioned on reliable observations, rather than merely recombining existing candidates.

Search performance under practical verification. We replace Oracle verification with DINO-centroid to evaluate search without ground-truth target views, and additionally include IPR-style Refinement as a verifier-free partial-refinement baseline. As shown in Fig. 8, independent view composition loses its Oracle advantage under practical verification, as individually selected views may lack cross-view consistency. IPR-style Refinement performs random partial regeneration without explicitly identifying unreliable viewpoints, while E-R Search uses DINO-centroid to combine coherent set-level selection with targeted refinement. Reliable views are retained as observations to guide the regeneration of uncertain views. These results show

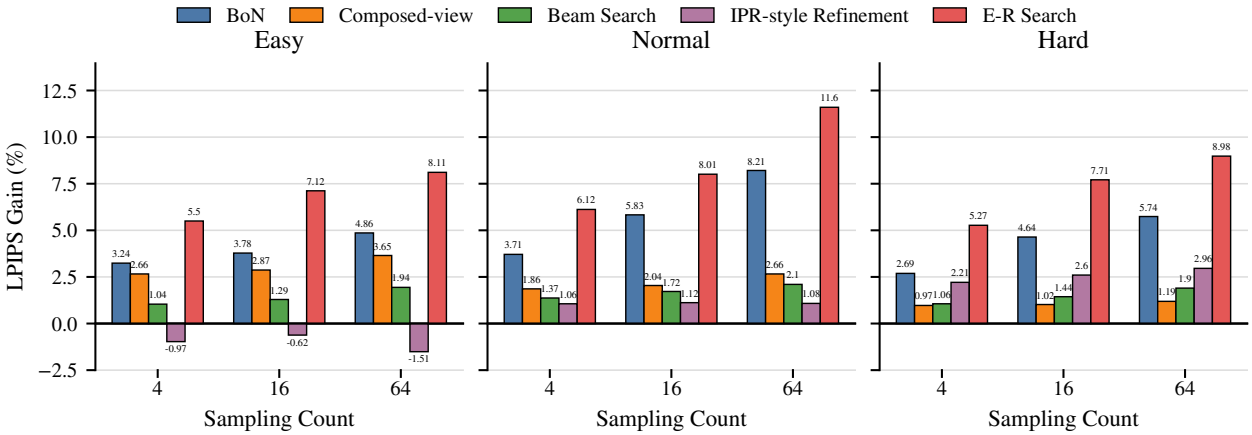


Figure 8: **Search performance under practical verification.** BoN, View-composed BoN, Beam Search, and E-R Search use DINO-centroid for candidate selection, whereas IPR-style Refinement is verifier-free and performs random partial refinement.

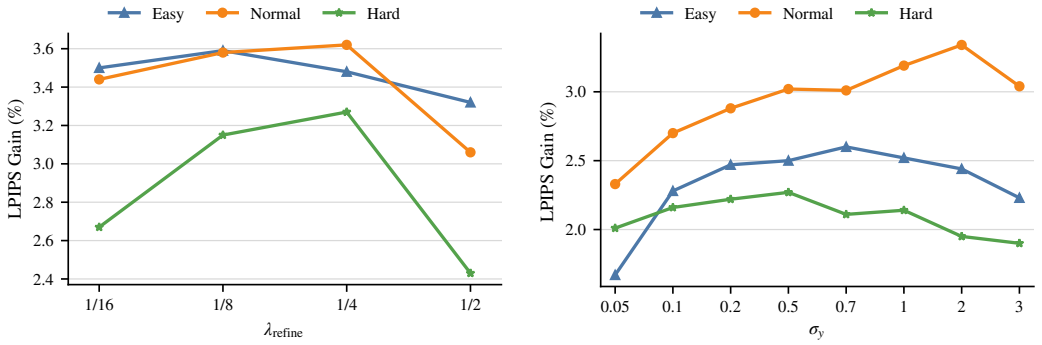


Figure 9: **Ablation studies on E-R Search.** Left: effect of exploration–refinement allocation λ_{refine}. Right: effect of observation noise σ_y.

that, under matched inference budgets, verifier-guided targeted refinement provides stronger gains than both independent view composition and verifier-free random partial refinement.

Exploration–Refinement Trade-off and Observation Uncertainty. As shown in Fig. 9, we analyze two key factors in E-R Search: the allocation of test-time computation between exploration and refinement, and the strength of observation conditioning. For the former, we vary the refinement ratio $\lambda_{\text{refine}} = R/(K+R)$ under a fixed budget, where larger values allocate more computation to refinement and smaller values favor exploration. For the latter, we vary the observation noise σ_y: smaller values impose stronger consistency with the observations, while larger values allow more flexible refinement. More details are provided in Appendix E.6.

4 Conclusion

We systematically studied test-time scaling for single-image-conditioned multi-view diffusion models through generation, verification, and search. Our experiments across multiple evaluated models show that joint multi-view generation exhibits stronger scaling behavior than independent per-view generation, while also

introducing a tension between preserving cross-view dependencies and exploiting view-level improvement opportunities. To address verification under this structure, we introduced DINO-centroid, a lightweight cross-sample consensus verifier whose representation becomes increasingly accurate as the sampling budget grows. Building on these observations, we proposed E-R Search, which combines set-level exploration with targeted view-level refinement formulated as a diffusion inverse problem and improves reconstruction quality under matched inference budgets. However, our approach remains limited by the accuracy of practical verifiers: refinement depends on the quality of verifier-selected observations, and inaccurate view identification may provide suboptimal guidance for regeneration. Our experiments also focus on single-image-conditioned multi-view generation, leaving broader structured generation tasks for future study.

Overall, our results show that effective test-time scaling for multi-view generation requires jointly considering dependencies in generation, verification granularity, and search. More broadly, they suggest that explicitly accounting for dependencies among output components can be important when extending test-time scaling to structured outputs.

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A DINO-centroid Implementation Details

A.1 Feature Extraction and View-specific Centroids

We use a frozen pretrained DINOv2 encoder ϕ_{DINO}. For view v in candidate set i, we extract and normalize its feature:

$$\mathbf{h}_i^{(v)} = \phi_{\text{DINO}}(x_i^{(v)}), \qquad \mathbf{z}_i^{(v)} = \mathbf{h}_i^{(v)} / \|\mathbf{h}_i^{(v)}\|_2. \tag{3}$$

For K independently generated sets, a separate normalized centroid is constructed for each viewpoint, without mixing views from different poses or objects:

$$\bar{\mathbf{z}}^{(v)} = \frac{\sum_{i=1}^{K} \mathbf{z}_i^{(v)}}{\left\| \sum_{i=1}^{K} \mathbf{z}_i^{(v)} \right\|_2}. \tag{4}$$

A.2 Set Selection and Uncertain-view Identification

The same view-specific similarity scores support both selection tasks:

$$s_i^{(v)} = \left(\mathbf{z}_i^{(v)}\right)^\top \bar{\mathbf{z}}^{(v)}. \tag{5}$$

$$R_{\text{set}}(\mathbf{X}_i) = \frac{1}{V} \sum_{v=1}^{V} s_i^{(v)}. \tag{6}$$

The set-level winner and the per-view winners are $i^* = \arg\max_i R_{\text{set}}(\mathbf{X}_i)$ and $i_v^* = \arg\max_i s_i^{(v)}$, respectively. We retain the complete set $\mathbf{X}^* = \mathbf{X}_{i^*}$ and identify views whose selected-set candidate is not the highest-ranked candidate for that pose:

$$\mathcal{V}_{\text{unc}} = \{v : i_v^* \neq i^*\}. \tag{7}$$

The remaining viewpoints form the reliable set $\mathcal{V}_{\text{rel}} = \mathcal{V} \setminus \mathcal{V}_{\text{unc}}$. This relative criterion requires neither a hand-set score threshold nor a fixed number of uncertain views. "Reliable" denotes a view selected by the verifier, not a guarantee of ground-truth accuracy. The views are not independently recomposed; the reliable views instead become observations for conditional refinement.

A.3 Computational Complexity

For K samples, V viewpoints, and feature dimension d, centroid construction and scoring together cost $\mathcal{O}(KVd)$ after feature extraction. Selection and view identification reuse the same scores and require no 3D reconstruction or dense cross-view correspondence estimation.

B Correlation Between DINO Similarity and Reconstruction Quality

We investigate whether DINO representations provide meaningful signals for evaluating reconstruction quality. For diagnostic analysis, we compute the cosine similarity between the DINO features of each generated view and its corresponding ground-truth view, and compare it with PSNR, SSIM, and LPIPS.

As shown in Fig. 10, DINO similarity exhibits moderate rank correlations with reconstruction metrics, with correlation magnitudes of approximately 0.4. Higher DINO similarity generally corresponds to higher PSNR and SSIM and lower LPIPS. These results suggest that DINO representations capture information relevant to reconstruction quality, supporting their use as a feature space for candidate verification.

Importantly, ground-truth views are used only for this diagnostic analysis and are not required by DINO-centroid during practical verification.

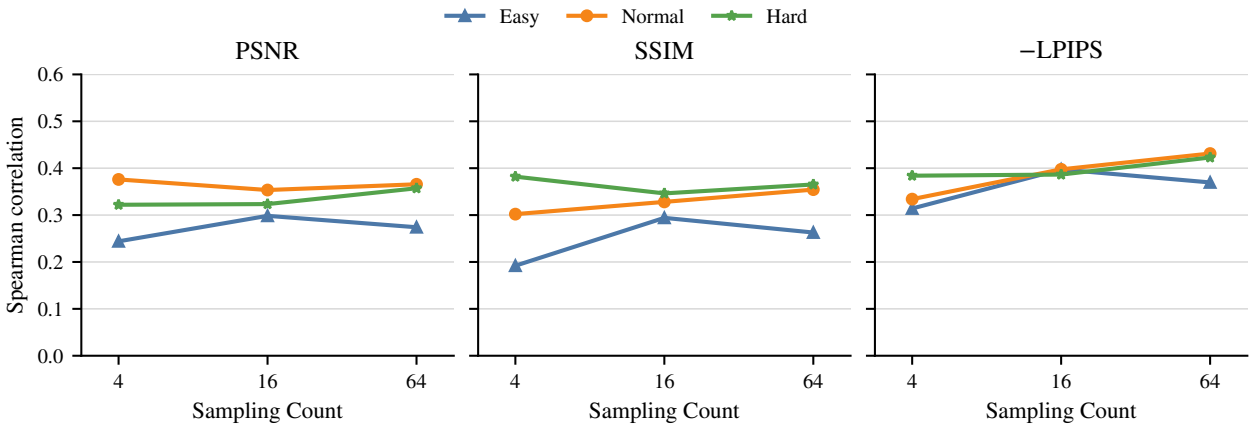


Figure 10: **Correlation between DINO similarity and reconstruction quality.** Ground-truth DINO similarity is compared with PSNR, SSIM, and LPIPS using Spearman correlations. Ground-truth information is used only for diagnostic analysis.

C Ambiguity in Multi-View Generation

A single reference image does not uniquely determine an object's occluded geometry or appearance. Figure 11 illustrates how the same input can yield distinct yet plausible multi-view hypotheses, particularly at viewpoints farther from the reference view.

D Comparison of Multi-View Verifiers

We provide additional details on the verification signals, score aggregation strategies, and computational costs of the multi-view verifiers considered in this work. We consider four practical verifiers: Aesthetic Score, Cross-view Matching, MEt3R, and our proposed DINO-centroid. Oracle verification is included as an analytical reference and requires access to ground-truth target views.

D.1 Verification Signals and Score Aggregation

Oracle verification. Oracle verification evaluates generated views against their corresponding ground-truth images using PSNR, SSIM, or LPIPS. For set-level verification, we average the reconstruction metric

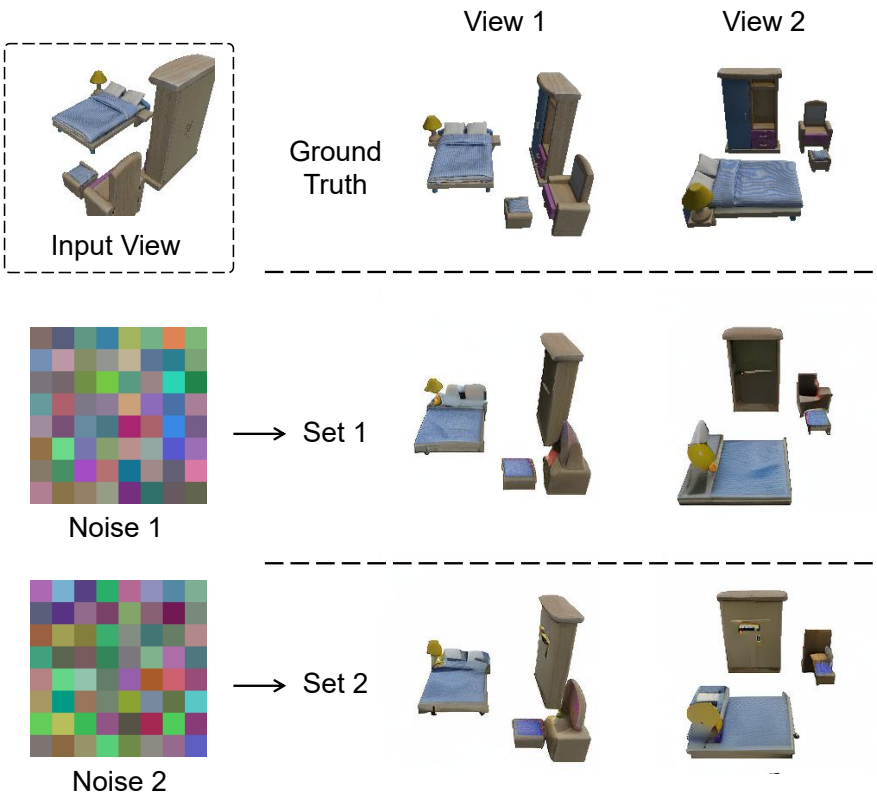


Figure 11: **Ambiguity in single-image-conditioned multi-view generation.** The same reference image leads to two plausible hypotheses: the wardrobe doors are open in Set 1 and closed in Set 2. The difference is more visible in View 2 than in View 1.

across all target viewpoints. PSNR and SSIM are maximized, whereas LPIPS is minimized. Oracle verification provides a reference for analyzing the potential of test-time search but is not available in practical deployment.

Aesthetic Score. Aesthetic Score evaluates the perceptual quality of individual generated images without requiring ground-truth target views. Let $R_{\text{aes}}(x_i^{(v)})$ denote the aesthetic score of viewpoint v in sample i. For set-level verification, we average the scores across all viewpoints:

$$R_{\text{set}}^{\text{aes}}(\mathbf{X}_i) = \frac{1}{V}\sum_{v=1}^{V} R_{\text{aes}}(x_i^{(v)}). \tag{8}$$

This aggregation enables an image-level quality estimator to evaluate complete multi-view samples, although it does not explicitly measure consistency between different viewpoints.

Cross-view Matching. Cross-view Matching evaluates 2D consistency by measuring image correspondences between pairs of generated views.

Rather than evaluating all possible view pairs, we adopt a cyclic adjacent-view evaluation protocol. For a multi-view sample containing $V = 16$ viewpoints, we evaluate the 16 adjacent pairs

$$\mathcal{P} = \{(v, (v \bmod V) + 1)\}_{v=1}^{V}, \tag{9}$$

including the pair connecting the last viewpoint to the first.

Let $s_{\text{match}}(x_i^{(v)}, x_i^{(u)})$ denote the number of matched feature points between two views, with a larger number of matched points indicating better consistency. The set-level score is computed as

$$R_{\text{set}}^{\text{match}}(\mathbf{X}_i) = \frac{1}{V}\sum_{(v,u)\in\mathcal{P}} s_{\text{match}}\left(x_i^{(v)}, x_i^{(u)}\right). \tag{10}$$

This protocol evaluates local cross-view consistency while avoiding the computational cost of exhaustive pairwise comparisons.

MEt3R. MEt3R evaluates geometric consistency between a pair of images. Its score ranges from 0 to 1, with lower values indicating higher cross-view consistency.

We adopt the same cyclic adjacent-view protocol as Cross-view Matching. The set-level MEt3R score is

$$E_{\text{set}}^{\text{MEt3R}}(\mathbf{X}_i) = \frac{1}{V}\sum_{(v,u)\in\mathcal{P}} \text{MEt3R}\left(x_i^{(v)}, x_i^{(u)}\right). \tag{11}$$

For consistency with our score-maximization formulation, we define the corresponding verification reward as

$$R_{\text{set}}^{\text{MEt3R}}(\mathbf{X}_i) = -E_{\text{set}}^{\text{MEt3R}}(\mathbf{X}_i). \tag{12}$$

Unlike image-level quality estimators, MEt3R explicitly evaluates geometric compatibility between generated viewpoints.

DINO-centroid. Our DINO-centroid verifier constructs a view-specific consensus representation from independently generated multi-view samples.

Given K samples, let $\mathbf{z}_i^{(v)}$ denote the ℓ_2-normalized DINO feature of viewpoint v in sample i. The view-specific centroid is computed as

$$\bar{\mathbf{z}}^{(v)} = \frac{\sum_{i=1}^{K} \mathbf{z}_i^{(v)}}{\left\| \sum_{i=1}^{K} \mathbf{z}_i^{(v)} \right\|_2}. \tag{13}$$

Each generated view is evaluated by its cosine similarity to the corresponding centroid:

$$R_{\text{view}}^{\text{DINO}}(x_i^{(v)}) = (\mathbf{z}_i^{(v)})^\top \bar{\mathbf{z}}^{(v)}. \tag{14}$$

The set-level score is obtained by averaging the view-level scores:

$$R_{\text{set}}^{\text{DINO}}(\mathbf{X}_i) = \frac{1}{V} \sum_{v=1}^{V} R_{\text{view}}^{\text{DINO}}(x_i^{(v)}). \tag{15}$$

Unlike Aesthetic Score, DINO-centroid exploits information shared across multiple generations of the same object. Unlike Cross-view Matching and MEt3R, it does not require explicit cross-view correspondence estimation or 3D reconstruction.

The resulting view-level scores also enable the identification of potentially suboptimal views for subsequent refinement.

D.2 Computational Complexity and Runtime

We compare the computational costs of different verification strategies in terms of the number of candidate samples K and viewpoints V.

Let C_{aes}, C_{match}, C_{MEt3R}, and C_{DINO} denote the costs of one aesthetic evaluation, one image-pair matching evaluation, one MEt3R evaluation, and one DINO feature extraction, respectively. Let d denote the dimensionality of the DINO feature representation.

Aesthetic Score requires one evaluation per generated view, giving a total complexity of $\mathcal{O}(KVC_{\text{aes}})$.

Exhaustive pairwise consistency evaluation would require $\mathcal{O}(KV^2)$ image-pair evaluations. By evaluating only cyclically adjacent views, our protocol reduces this to KV pair evaluations, yielding complexities of $\mathcal{O}(KVC_{\text{match}})$ and $\mathcal{O}(KVC_{\text{MEt3R}})$ for Cross-view Matching and MEt3R, respectively.

DINO-centroid requires feature extraction for all KV generated views. Given the extracted features, centroid construction and candidate scoring together require $\mathcal{O}(KVd)$ operations. Its overall complexity is therefore

$$\mathcal{O}(KVC_{\text{DINO}} + KVd). \tag{16}$$

Importantly, the same DINO features and view-specific centroids are reused for set-level selection and uncertain-view identification, without additional feature extraction.

Table 1 reports the measured runtime per multi-view sample. DINO-centroid achieves the lowest runtime among the compared verifiers, while Cross-view Matching and MEt3R incur higher computational costs due to their pairwise consistency evaluations.

Since DINO-centroid constructs its centroids from multiple samples, its reported per-sample runtime should be interpreted as an amortized cost over the evaluated candidate pool.

Table 1: **Comparison of multi-view verifiers.** Oracle requires ground-truth views and is included only as an analytical reference. Runtime is reported per multi-view sample.

Verifier	Evaluation signal	Time (s)
Oracle (PSNR, SSIM, LPIPS)	Ground-truth reconstruction	0.10
Aesthetic Score	Image quality	0.12
Cross-view Matching	2D cross-view consistency	2.28
MEt3R	3D geometric consistency	1.05
DINO-centroid	Cross-sample feature consensus	**0.07**

E Details of E-R Search

E.1 From Verification to Observations

Given K independently generated multi-view sets, we first select $\mathbf{X}^*$ using the set-level verifier. In our main experiments, this verifier is DINO-centroid (Appendix A), which assigns a score to every generated view and averages the view-level scores across viewpoints to obtain the corresponding set-level score.

We further use these view-level DINO-centroid scores to determine which views in $\mathbf{X}^*$ are preserved during refinement. Specifically, for each viewpoint v, we compare the selected view $x^{*(v)}$ with the other $K-1$ candidates generated for the same viewpoint. If $x^{*(v)}$ achieves the highest DINO-centroid score among these candidates, we regard it as a *reliable* view; otherwise, it is treated as *uncertain*. This partitions the viewpoints into $\mathcal{V}_{\text{rel}}$ and $\mathcal{V}_{\text{unc}}$.

Thus, in the practical setting, the observations used by E-R Search are not ground-truth images, but verifier-selected views from actually generated multi-view samples. The verifier therefore plays two roles: it selects a complete multi-view hypothesis and identifies which views within that hypothesis should be preserved or regenerated.

For Oracle experiments, ground-truth reconstruction metrics are used only to make the corresponding selection decisions; the ground-truth target views are not provided as observations during ordinary refinement. We separately conduct a controlled observation experiment in which ground-truth views are explicitly supplied as observations to study the effect of observation quality.

E.2 Inverse Problem Formulation

Let $\mathcal{E}$ denote the pretrained model's image encoder and

$$\mathbf{Z}_{\text{ref}} = \{\mathcal{E}(x^{*(v)})\}_{v=1}^{V} \tag{17}$$

the latent representation of the selected multi-view set. We define a view mask

$$M_v = \begin{cases} 1, & v \in \mathcal{V}_{\text{rel}}, \\ 0, & v \in \mathcal{V}_{\text{unc}}, \end{cases} \tag{18}$$

which is broadcast over all channels and spatial locations of each view. The reliable-view observations are then

$$\mathbf{y} = M \odot \mathbf{Z}_{\text{ref}}. \tag{19}$$

We formulate refinement as a masked inverse problem,

$$\mathbf{y} = M \odot \mathbf{Z}_0 + M \odot \mathbf{n}, \qquad \mathbf{n} \sim \mathcal{N}(0, \sigma_y^2 I), \tag{20}$$

where σ_y represents the assumed uncertainty of the generated observations. For $\sigma_y > 0$, the likelihood over observed entries is

$$p(\mathbf{y} \mid \mathbf{Z}_0, M) \propto \exp\left(-\frac{\|M \odot \mathbf{Z}_0 - \mathbf{y}\|_2^2}{2\sigma_y^2}\right), \tag{21}$$

which yields the posterior

$$p_\theta(\mathbf{Z}_0 \mid \mathbf{y}, M, c) \propto p(\mathbf{y} \mid \mathbf{Z}_0, M) p_\theta(\mathbf{Z}_0 \mid c). \tag{22}$$

Smaller σ_y imposes stronger agreement with the preserved views; in the limit $\sigma_y \to 0$, the observations become hard constraints,

$$M \odot \mathbf{Z}_0 = \mathbf{y}. \tag{23}$$

Because the diffusion prior jointly models the complete multi-view set, the uncertain views can be regenerated while remaining conditioned on the preserved views through the joint denoising process.

E.3 Sampling and Conditioning

We keep the pretrained multi-view diffusion model frozen. Starting from a noised version of the selected latent, we initialize

$$\mathbf{Z}_{t_s} = \sqrt{\bar{\alpha}_{t_s}}\, \mathbf{Z}_{\text{ref}} + \sqrt{1 - \bar{\alpha}_{t_s}}\, \boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(0, I), \tag{24}$$

where t_s is determined by the refinement strength.

At each reverse step t, the model predicts

$$\hat{\boldsymbol{\epsilon}}_t = \epsilon_\theta(\mathbf{Z}_t, t, c), \tag{25}$$

from which the clean latent estimate is obtained as

$$\hat{\mathbf{Z}}_0^{(t)} = \frac{\mathbf{Z}_t - \sqrt{1 - \bar{\alpha}_t}\, \hat{\boldsymbol{\epsilon}}_t}{\sqrt{\bar{\alpha}_t}}. \tag{26}$$

For reference, the standard DDIM transition is

$$\mathbf{Z}_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\, \hat{\mathbf{Z}}_0^{(t)} + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2}\, \hat{\boldsymbol{\epsilon}}_t + \sigma_t \boldsymbol{\xi}_t, \qquad \boldsymbol{\xi}_t \sim \mathcal{N}(0, I). \tag{27}$$

Posterior-DDIM (Jiao et al., 2026) modifies this reverse process by incorporating the measurement $\mathbf{y}$ on the entries selected by M. At each applicable denoising step, the current clean estimate is corrected toward the observed reliable-view latents, with the correction strength determined by the diffusion noise level and the assumed observation uncertainty σ_y. No direct measurement constraint is applied to entries corresponding to $\mathcal{V}_{\text{unc}}$; these views are therefore regenerated under the joint multi-view diffusion prior while being coupled to the preserved views through the model.

In practice, we additionally use an SNR-dependent switching rule to determine when the measurement signal is injected. Rather than enforcing the generated observations throughout the entire reverse trajectory, we activate the posterior correction only once the relative signal-to-noise levels make the observation sufficiently informative. Before this point, sampling follows the ordinary reverse dynamics; afterward, the reliable-view observations are incorporated during denoising. We found that applying observation correction at inappropriate noise levels can cause excessive averaging between the model prediction and the measurement, leading to overly smooth or blurry reconstructions.

The conditioning image c remains the original input to the pretrained multi-view diffusion model throughout refinement, whereas $\mathbf{y}$ contains the verifier-selected reliable views used as additional observations. When the original input image is explicitly enforced as an observation, it replaces view 0 before encoding and view 0 is included in $\mathcal{V}_{\text{rel}}$. After the reverse process terminates, the resulting latent set is decoded to obtain the refined multi-view sample.

E.4 Exploration–Refinement Procedure

E-R Search first generates K independent complete multi-view sets and selects one using the set-level verifier. In the practical setting, DINO-centroid is then used as described above to divide the selected set into reliable and uncertain views. The reliable views are converted into observations, while the uncertain views are left unconstrained during Posterior-DDIM refinement.

We then perform R refinement trajectories. All trajectories share the same selected reference set and observation mask, but use independent sampling noise, thereby producing alternative completions for the uncertain views while preserving information from the reliable ones. The refined candidates are evaluated together with the explored candidates using the set-level verifier, and the highest-scoring complete set is returned as the final output.

To compare against Best-of-N under approximately matched inference budgets, we allocate

$$K + R = N. \tag{28}$$

A full refinement trajectory uses the same number of diffusion steps as a complete generation trajectory and therefore incurs approximately the same number of model evaluations. Verifier overhead is reported separately.

E.5 Algorithmic Procedure

Algorithm 1 summarizes E-R Search. The method first generates K independent multi-view candidates, selects a complete sample using DINO-centroid, and partitions its views into reliable and uncertain subsets. It then performs R conditional refinement trajectories, preserving the reliable views as observations while regenerating the uncertain views. Within each refinement trajectory, we use SNR-switched Posterior DDIM. At each reverse step, the reliable views follow the measurement-conditioned update when the measurement SNR exceeds the diffusion SNR, and otherwise follow the standard diffusion prior update. The uncertain views are regenerated under the joint multi-view prior throughout the reverse process. Finally, the selected sample and all refined candidates are scored by the set-level verifier, and the highest-scoring result is returned.

E.6 Implementation Details

Unless otherwise specified, refinement uses the pretrained SyncDreamer model with $V = 16$ views, 50 DDIM steps, classifier-free guidance scale 2.0, and camera elevation 30°. We use refinement strength 1.0 for full-resampling experiments. The default refinement ratio is $\lambda_{\text{refine}} = 1/4$, and the default observation uncertainty is $\sigma_y = 0.1$.

F Qualitative Case Studies of Verification-Guided Refinement

We provide qualitative case studies to illustrate how E-R Search uses verification to guide targeted view-level refinement. For each example, the leftmost image is the input view, and the four rows show the ground-truth target views, an unselected generated set, the verifier-selected set before refinement, and the final refined set after Posterior-DDIM sampling.

In the selected set, views outlined by red boxes are identified as uncertain and selected for regeneration, whereas views outlined by green boxes are retained as reliable observations during refinement. Here, reliable observations denote views retained according to the DINO-centroid verifier, rather than views guaranteed to be ground-truth correct. The refined set is obtained by regenerating the red-boxed views while conditioning on the preserved green-boxed views.

Algorithm 1 Exploration–Refinement Search with SNR-Switched Posterior DDIM

Require: Input image c, joint multi-view diffusion model p_θ, sampling budget N, exploration budget K, refinement budget R, measurement uncertainty σ_y, verifier R_{DINO}

Ensure: Refined multi-view sample $\mathbf{X}^*$

$R \leftarrow N - K$

{Stage 1: Exploration}

for $i = 1, \ldots, K$ **do**

  $\mathbf{X}_i \sim p_\theta(\mathbf{X} \mid c, \Pi)$

end for

{Stage 2: Verification}

Compute view-specific DINO centroids $\{\bar{\mathbf{z}}^{(v)}\}_{v=1}^{V}$ from $\{\mathbf{X}_i\}_{i=1}^{K}$

Compute view-level scores $s_i^{(v)} = (\mathbf{z}_i^{(v)})^\top \bar{\mathbf{z}}^{(v)}$

Compute set-level scores $S_i = \frac{1}{V} \sum_{v=1}^{V} s_i^{(v)}$

$i^* \leftarrow \arg\max_i S_i, \quad \mathbf{X}_{\mathrm{ref}} \leftarrow \mathbf{X}_{i^*}$

$\mathcal{V}_{\mathrm{unc}} \leftarrow \{v : \arg\max_i s_i^{(v)} \neq i^*\}$

$\mathcal{V}_{\mathrm{rel}} \leftarrow \{1, \ldots, V\} \setminus \mathcal{V}_{\mathrm{unc}}$

{Stage 3: Conditional Refinement}

Encode $\mathbf{X}_{\mathrm{ref}}$ as $\mathbf{Z}_{\mathrm{ref}}$

$\mathcal{C} \leftarrow \{\mathbf{X}_{\mathrm{ref}}\}$

for $r = 1, \ldots, R$ **do**

  Sample $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$

  Initialize $\mathbf{Z}_{t_s} = \sqrt{\bar{\alpha}_{t_s}} \mathbf{Z}_{\mathrm{ref}} + \sqrt{1 - \bar{\alpha}_{t_s}} \boldsymbol{\epsilon}$

  for $t = t_s, \ldots, 1$ **do**

    Predict noise $\hat{\boldsymbol{\epsilon}}_t \leftarrow \epsilon_\theta(\mathbf{Z}_t, t, c)$

    Compute the prior DDIM update $\mathbf{Z}_{t-1}^{\mathrm{prior}}$

    $\mathrm{SNR}_{\mathrm{meas}} \leftarrow 1/\sigma_y$

    $\mathrm{SNR}_{\mathrm{diff}}(t-1) \leftarrow \sqrt{\bar{\alpha}_{t-1}/(1 - \bar{\alpha}_{t-1})}$

    $\mathbf{Z}_{t-1} \leftarrow \mathbf{Z}_{t-1}^{\mathrm{prior}}$

    if $\mathrm{SNR}_{\mathrm{meas}} > \mathrm{SNR}_{\mathrm{diff}}(t-1)$ **then**

      Sample $\boldsymbol{\xi} \sim \mathcal{N}(0, \mathbf{I})$

      $\mathbf{Z}_{t-1}^{\mathrm{obs}} \leftarrow \sqrt{\bar{\alpha}_{t-1}}\, \mathbf{Z}_{\mathrm{ref}} + \sqrt{1 - \bar{\alpha}_{t-1} - \bar{\alpha}_{t-1}\sigma_y^2}\, \boldsymbol{\xi}$

      Replace $\mathbf{Z}_{t-1}^{(\mathcal{V}_{\mathrm{rel}})} \leftarrow (\mathbf{Z}_{t-1}^{\mathrm{obs}})^{(\mathcal{V}_{\mathrm{rel}})}$

    end if

  end for

  Decode $\widetilde{\mathbf{X}}_r \leftarrow \mathcal{D}(\mathbf{Z}_0)$

  $\mathcal{C} \leftarrow \mathcal{C} \cup \{\widetilde{\mathbf{X}}_r\}$

end for

{Final Selection}

$\mathbf{X}^* \leftarrow \arg\max_{\mathbf{X} \in \mathcal{C}} R_{\mathrm{DINO}}(\mathbf{X})$

return $\mathbf{X}^*$

These examples illustrate the complementary roles of set-level selection and view-level refinement in E-R Search. At the set level, DINO-centroid selects a complete multi-view hypothesis instead of independently composing views from different samples. At the view level, the same verification signal identifies which views should be preserved as reliable observations and which should be regenerated. This allows E-R Search to refine uncertain viewpoints while maintaining conditioning from the selected multi-view hypothesis.

G Experimental Setup and Reproducibility

G.1 Datasets and Evaluation Protocol

Our multi-view experiments use Google Scanned Objects (GSO) (Downs et al., 2022), with one reference image and 16 target views per object. We rank objects by single-sample LPIPS and construct fixed Easy, Normal, and Hard subsets of 100 objects each: Easy contains the 100 objects with the lowest LPIPS, Hard contains the 100 with the highest LPIPS, and Normal contains 100 objects randomly sampled from the remaining objects. All methods use the same objects, input images, camera poses, and target images within each comparison.

G.2 Evaluation Metrics

We average PSNR, SSIM, and LPIPS over the evaluated target views; higher PSNR/SSIM and lower LPIPS indicate better reconstruction. In the controlled ground-truth observation experiment, only unobserved target views are evaluated: the input view and all provided ground-truth observations are excluded. For a higher-is-better metric Q and for LPIPS, we report

$$\Delta Q(N) = \frac{Q_N - Q_1}{|Q_1|} \times 100\%, \qquad \Delta\mathrm{LPIPS}(N) = \frac{\mathrm{LPIPS}_1 - \mathrm{LPIPS}_N}{\mathrm{LPIPS}_1} \times 100\%. \tag{29}$$

Positive values indicate improvements in both cases. We also evaluate cross-view consistency using MEt3R and 2D image matching; MEt3R is a lower-is-better consistency measure.

G.3 Generation Models and Sampling Configuration

SyncDreamer is our joint-generation model, while Zero-1-to-3 serves as the per-view baseline. Unless otherwise specified, SyncDreamer uses 16 views, 50 DDIM steps, guidance scale 2.0, and elevation 30°. Posterior-DDIM uses the same frozen SyncDreamer model; its formulation and refinement parameters are given in Appendix E.

G.4 Search Budget and Computational Cost

We evaluate all search methods under matched computational budgets, measured by the number of denoising function evaluations (NFEs), with $N \in \{4, 16, 64\}$ denoising-trajectory equivalents. Best-of-N uses N independent generation trajectories. E-R Search allocates K trajectories to independent exploration and R trajectories to full refinement, with $K + R = N$. Beam Search uses configurations $(B, E) = (2, 2), (4, 4), (8, 8)$ under the corresponding NFE budgets.

For IPR-style Refinement, one trajectory is used to generate the initial multi-view sample, followed by $N - 1$ iterative partial-refinement trajectories. At each refinement iteration, a random subset of views is selected for regeneration while the remaining views are retained as observations. Each Posterior-DDIM refinement uses the same number of denoising steps as a complete generation and is therefore counted as one trajectory-equivalent unit of computation.

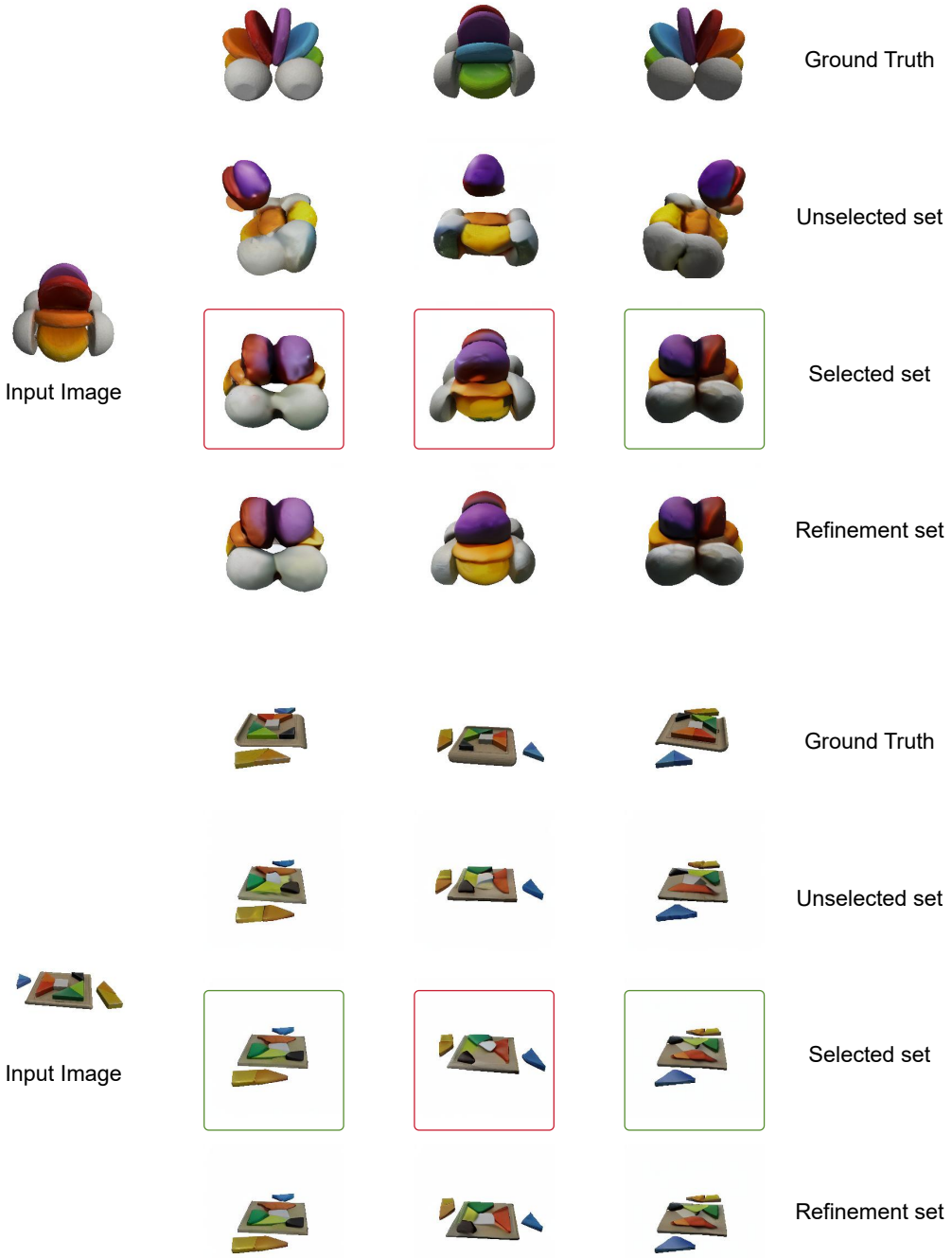


Figure 12: **Qualitative case studies of verification-guided targeted refinement.** In each example, the leftmost image is the input view. The four rows show the ground-truth target views, an unselected generated set, the verifier-selected set before refinement, and the final refined set after Posterior-DDIM sampling. In the selected set, red boxes indicate uncertain views selected for regeneration, while green boxes indicate reliable observations retained according to the DINO-centroid verifier. The refined set is obtained by regenerating the red-boxed views while conditioning on the preserved green-boxed views.

Since verifying a complete multi-view sample takes less than 1% of the time required to generate a complete multi-view sample, we neglect verification overhead when comparing computational budgets.

G.5 IPR-style Refinement Baseline

We additionally adapt Iterative Partial Refinement (IPR) to joint multi-view generation as a recent verifier-free test-time refinement baseline. The original IPR framework performs iterative partial regeneration by randomly selecting regions for resampling while conditioning on the remaining generated content. Since the output components in our setting are camera views rather than spatial image regions, we instantiate the same principle at the view level.

Starting from an initially generated multi-view sample, at each refinement iteration we randomly select a subset of target views for regeneration, while retaining the remaining views as observations. The selected views are regenerated using the same Posterior-DDIM conditional sampling procedure described in Appendix E, but view selection is entirely random and does not use DINO-centroid or any other verification signal. A new random subset is sampled at each iteration.

To match the scale of partial regeneration used by E-R Search, the number of randomly selected views is matched to the average number of uncertain views identified by DINO-centroid, corresponding to approximately 7.3 out of 16 target views in our experiments. IPR-style Refinement is therefore verifier-free: neither the views to regenerate nor the intermediate refinement results are selected using DINO-centroid.

We match the nominal inference budgets used by the other search methods. One complete generation or one full Posterior-DDIM refinement trajectory counts as one trajectory-equivalent unit of computation.

G.6 Reproducibility

We use fixed evaluation splits and camera protocols, frozen pretrained models, and matched settings within each comparison. Source code, evaluation scripts, and configurations will be provided through the anonymous repository identified in the Reproducibility Statement.

H Additional Analysis of Test-Time Scaling Across Generation Paradigms

Our main experiments compare Zero-1-to-3 and SyncDreamer to investigate the effect of generation paradigms on test-time scaling (TTS). While SyncDreamer exhibits stronger Oracle scaling, these two models also differ in architecture, training data, and learned generative distributions. Therefore, the observed difference cannot be attributed solely to joint multi-view modeling.

To further examine whether this trend generalizes across model families, we consider two additional per-view generation models, Zero123-XL (Liu et al., 2023) and Stable Zero123 (Stability AI, 2023), and two joint multi-view generation models, Zero123++ (Shi et al., 2023) and MV-Adapter (Huang et al., 2025), with results reported in Table 2.

For a controlled comparison, we evaluate all models on the same object set and target camera poses, using identical image preprocessing and evaluation metrics. For each model, we generate $N \in \{1, 4, 16, 64\}$ independent multi-view candidates and apply the same view-level Oracle selection. Specifically, the best candidate for each target viewpoint is selected according to its LPIPS distance to the corresponding ground-truth image. We report the relative LPIPS improvement over a single sample.

This evaluation isolates the comparison of scaling behavior from differences in verification and selection strategies.

Table 2: Oracle LPIPS improvements (%) over single-sample generation under view-level selection.

Paradigm	Model	LPIPS Gain (%) ↑		
		$N = 4$	$N = 16$	$N = 64$
Per-view	Zero123-XL	9.1	13.5	16.3
	Stable Zero123	9.6	13.9	16.9
Joint	Zero123++	10.2	14.7	17.8
	MV-Adapter	9.8	15.1	18.2

I Generalization to Another Joint Multi-View Backbone

The main experiments use SyncDreamer as the joint multi-view diffusion backbone. To examine whether the proposed verification and search strategies remain effective with a different joint generator, we further evaluate them on MV-Adapter (Huang et al., 2025), another representative joint multi-view generation model.

Due to the computational cost of the additional backbone experiments, we conduct this evaluation on the Normal subset using the same objects and evaluation protocol as in the main experiments. We consider matched sampling budgets $N \in \{4, 16, 64\}$. Ground-truth target views are used only for evaluation unless otherwise specified, and we report relative LPIPS improvements over single-sample generation.

Verification. We first examine whether cross-sample feature consensus remains effective when the underlying joint generator is changed. Table 3 compares the practical verifiers considered in the main paper under Best-of-N selection. DINO-centroid consistently achieves the strongest selection performance across the evaluated sampling budgets, providing additional evidence that cross-sample feature consensus is not specific to SyncDreamer.

Table 3: **Verification performance on MV-Adapter.** Results are reported on the Normal subset. LPIPS improvements (%) over single-sample generation under Best-of-N selection are shown; higher is better.

Verifier	$N = 4$	$N = 16$	$N = 64$
Aesthetic Score	2.96	2.07	1.99
Cross-view Matching	1.45	0.98	0.30
MEt3R	3.58	3.26	1.80
DINO-centroid	**4.06**	**5.72**	**7.98**

Search. We next examine whether the exploration–refinement strategy remains effective with MV-Adapter. Table 4 compares Best-of-N, independent view composition, Beam Search, IPR-style Refinement, and E-R Search under matched inference budgets. BoN, Composed-view, Beam Search, and E-R Search use DINO-centroid for practical verification, whereas IPR-style Refinement is verifier-free and performs random partial refinement. E-R Search consistently outperforms the alternative search strategies across all evaluated budgets, showing the same overall trend as with SyncDreamer.

Overall, the results on the Normal subset show that both DINO-centroid verification and E-R Search remain effective when applied to MV-Adapter. Together with the SyncDreamer results, these experiments provide additional evidence that the proposed verification and search strategies are not specific to a single joint multi-view diffusion backbone.

Table 4: **Search performance on MV-Adapter.** Results are reported on the Normal subset. LPIPS improvements (%) over single-sample generation are reported under matched inference budgets; higher is better. BoN, Composed-view, Beam Search, and E-R Search use DINO-centroid, whereas IPR-style Refinement is verifier-free.

Search Method	$N = 4$	$N = 16$	$N = 64$
Best-of-N	4.06	5.72	7.98
Composed-view	2.28	2.61	2.93
Beam Search	1.25	1.64	2.25
IPR-style Refinement	0.92	0.80	0.87
E-R Search	**5.87**	**8.38**	**11.41**

J Cross-View Consistency of Search Methods

In addition to reconstruction quality, we evaluate how different search strategies affect cross-view consistency. We use MEt3R (Asim et al., 2025), a reference-free metric for measuring 3D consistency between generated views. Following our multi-view setting, we compute MEt3R between adjacent view pairs and average the scores over all views and objects. Lower MEt3R indicates better cross-view consistency.

Table 5 compares the five search strategies under different inference budgets and object difficulty levels. This evaluation complements the reconstruction metrics reported in the main paper by examining whether improvements obtained through test-time search preserve the dependencies among jointly generated views.

Table 5: Cross-view consistency of different search strategies measured by MEt3R (↓). Results are reported across different inference budgets and object difficulty levels.

Budget	**Search Method**	**Easy**	**Normal**	**Hard**
$N = 4$	Best-of-N	0.357	0.442	0.485
	Composed-view	0.368	0.488	0.571
	Beam Search	0.353	0.445	0.501
	IPR-style Refinement	0.355	**0.440**	0.508
	E-R Search	**0.350**	0.443	**0.472**
$N = 16$	Best-of-N	0.354	0.443	0.482
	Composed-view	0.374	0.509	0.591
	Beam Search	0.360	0.446	0.498
	IPR-style Refinement	0.351	0.439	0.506
	E-R Search	**0.348**	**0.438**	**0.476**
$N = 64$	Best-of-N	0.352	0.441	0.481
	Composed-view	0.372	0.508	0.599
	Beam Search	0.352	0.446	0.487
	IPR-style Refinement	0.349	0.436	0.503
	E-R Search	**0.344**	**0.434**	**0.468**

Overall, these results provide a complementary perspective on search performance beyond per-view reconstruction quality. Composed-view consistently yields substantially higher MEt3R scores, indicating that independently composing views from different samples can disrupt the cross-view dependencies established by joint generation. This degradation is particularly pronounced on harder objects and persists as the sampling budget increases.

In contrast, set-preserving strategies such as Best-of-N and Beam Search maintain substantially better

cross-view consistency, while IPR-style refinement provides comparable consistency under most settings. E-R Search achieves the lowest or near-lowest MEt3R across budgets and difficulty levels, with clearer advantages on hard objects and at larger budgets. Together with its reconstruction-quality improvements, these results suggest that E-R Search can selectively improve uncertain views while largely preserving, and in some settings further improving, the cross-view consistency of jointly generated samples.

K 3D Reconstruction Performance

We further evaluate whether the improvements in multi-view generation translate to downstream 3D reconstruction quality. Following SyncDreamer, we report Chamfer Distance and Volume IoU. We conduct this evaluation on the normal split and use the same reconstruction pipeline for all methods.

Table 6: **3D reconstruction performance.** Lower Chamfer Distance and higher Volume IoU indicate better reconstruction quality.

Method	Chamfer Dist. ↓	Volume IoU ↑
SyncDreamer	0.0274	0.537
Best-of-N	0.0265	0.551
Composed-view	0.0341	0.495
Beam Search	0.0268	0.540
IPR-style Refinement	0.0267	0.543
E-R Search	**0.0252**	**0.568**

As shown in Table 6, the 3D reconstruction results are consistent with multi-view evaluation. Best-of-N provides moderate improvements over the original SyncDreamer outputs, while Composed-view substantially degrades both Chamfer Distance and Volume IoU, suggesting that independently composing views can harm downstream 3D reconstruction despite improving individual viewpoints. E-R Search achieves the best performance on both metrics, indicating that preserving reliable views while selectively refining uncertain ones produces multi-view observations that are more suitable for 3D reconstruction.

L Ablation on Uncertain View Identification

Our refinement framework uses DINO-centroid to identify uncertain views for targeted regeneration. However, views that do not achieve the highest centroid similarity are not necessarily those with the largest reconstruction errors. We therefore investigate whether DINO-centroid provides a meaningful refinement signal beyond the benefits of conditional regeneration itself.

Experimental setup. We compare three view identification strategies under identical sampling budgets and refinement settings. All strategies share the same initial multi-view sample and regenerate the same number of views for each object.

For *Random*, we uniformly sample m distinct target views without replacement for regeneration. For *DINO-centroid*, we identify uncertain views whose scores are not the highest among the corresponding views of the initial candidate samples. For *Oracle*, we select the m views with the largest LPIPS distances to their corresponding ground-truth images.

For each strategy, the remaining views are treated as observations during Posterior-DDIM refinement. Random selection is repeated with multiple seeds to reduce the influence of stochastic variation.

Table 7: Ablation on uncertain-view identification under matched sampling budgets. We compare Random, DINO-centroid, and Oracle view selection. LPIPS improvements (%) are reported relative to single-sample generation.

Difficulty	N	**Random**	**DINO**	**Oracle**
Easy	4	3.85	5.50	6.12
	16	5.34	7.12	7.83
	64	6.42	8.11	8.79
Normal	4	4.45	6.12	6.79
	16	6.18	8.01	8.82
	64	9.02	11.60	12.05
Hard	4	3.52	5.27	6.05
	16	5.56	7.71	8.43
	64	6.67	8.98	9.65

Results. Table 7 compares different view identification strategies under matched refinement budgets. Random selection improves reconstruction quality over single-sample generation, confirming that conditional regeneration itself contributes to performance gains.

DINO-centroid consistently outperforms random selection across all sampling budgets and difficulty levels, demonstrating that cross-sample consensus provides an informative signal for identifying views that benefit from refinement. This improvement indicates that the effectiveness of E-R Search stems not only from conditional regeneration but also from the selection of which views to regenerate.

Oracle view selection achieves further improvements, revealing additional potential for more accurate uncertain-view identification. Together, these results demonstrate the complementary contributions of uncertain-view identification and conditional regeneration in our exploration–refinement framework.